ONE CAPTURE

Meal photo + context
day, meal band, zoom

Capture checks
invalid fields abstain

Scale anchor
60 mm ArUco marker on A4

SAM 2.1 + SigLIP 2
one mask; one dish label

invalid capture

unrecognized / stage error

accepted

unsupported

cannot_verify
invalid input, unknown dish, or stage error

MoGe-2 scene depth
accepted dish proceeds to mass estimation

out_of_scope
unsupported dish (for example, wheat)

PORTION MASS SOURCE

Marker found?

yes

no

MEASURED SCALE

Geometry mass
measured scale × area
× height × density

NO MARKER · MAPPED

WED mapped?
selected dish class

yes

no

Head available?
mapped dish only

yes

no

Learned mass
DINOv2-S head
coverage ×0.90

cannot_verify
head missing
or failed; no grams

NO MARKER · UNMAPPED

Size-prior geometry
prior grams · coverage ×0.60

Nutrient uncertainty
Monte Carlo propagation · 4,000 samples

Coverage-aware policy
PASS: all P ≥ 0.90, C ≥ 0.85
FAIL: any P ≤ 0.10, C ≥ 0.90

PASS

BORDERLINE

FAIL

Serialized response
verdict, reasons, failures